

Class 6 function homework

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Improving provided code

```
# Can you improve this analysis code?  
library(bio3d)  
s1 <- read.pdb("4AKE") # kinase with drug
```

Note: Accessing on-line PDB file

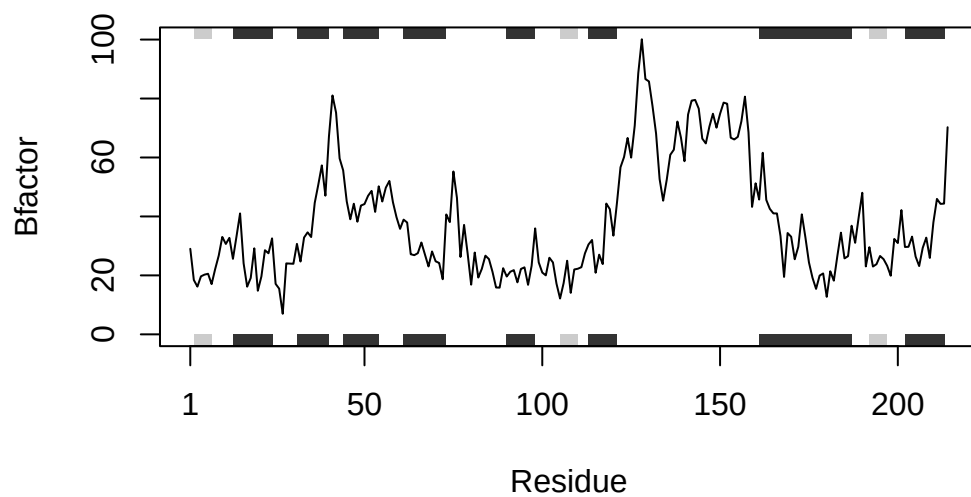
```
s2 <- read.pdb("1AKE") # kinase no drug
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

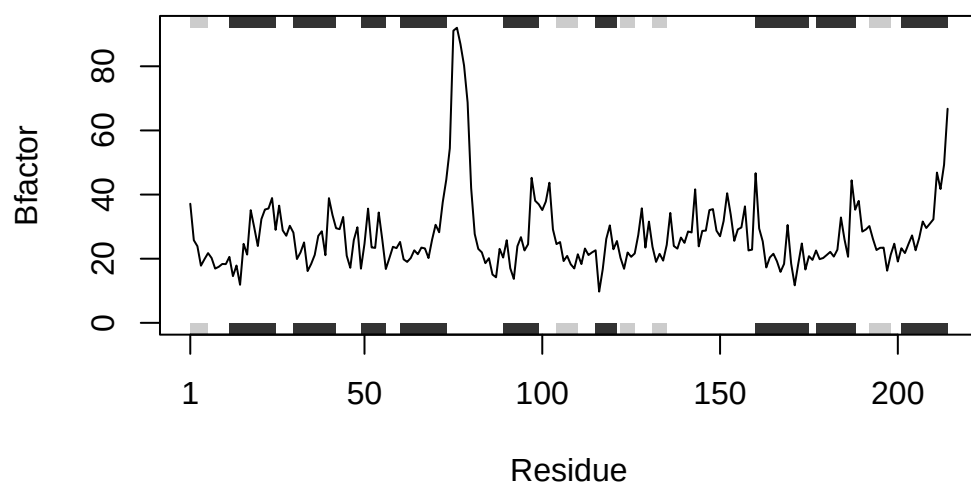
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file

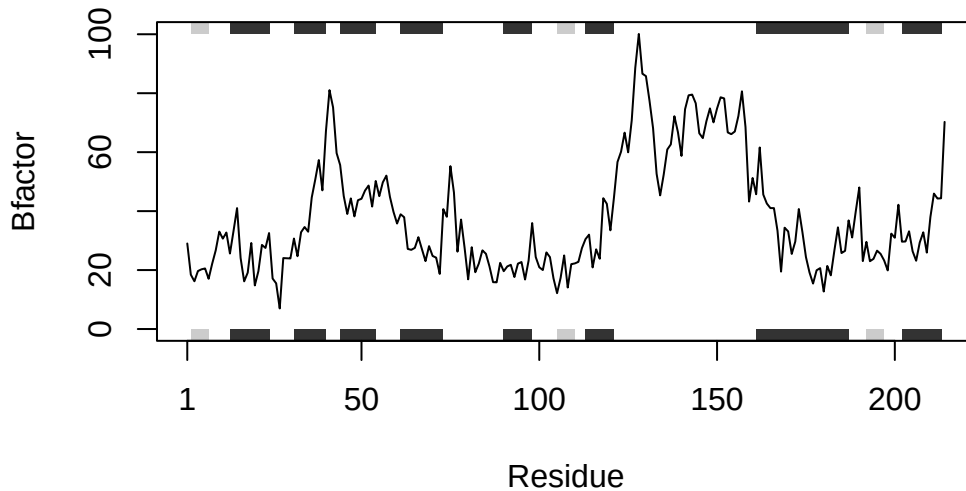
```
# This function accesses the online PDB database and retrieve the specified protein using its ID  
  
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")  
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")  
s3.chainA <- trim.pdb(s1, chain="A", elety="CA")  
# The function takes a 4-letter PDB identifier, a specific protein chain ID (e.g., "A"), and a selection string  
  
s1.b <- s1.chainA$atom$b  
s2.b <- s2.chainA$atom$b  
s3.b <- s3.chainA$atom$b  
# The input is a trimmed PDB object (e.g., s1.chainA) that contains atom information from a protein  
  
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



#The code takes a numeric vector of B-factors and the corresponding trimmed PDB object to pr

Write a function from the supplied code

```
library(bio3d)

# Function to analyze protein drug interactions using B-factors
analyze_protein_drug_interaction <- function(pdb_id, chain = "A", eley = "CA") {

  # Read protein structure
  pdb <- read.pdb(pdb_id)

  # Trim structure to selected chain and atom type
  pdb_chain <- trim.pdb(pdb, chain = chain, eley = eley)

  # Extract B-factors
  b_factors <- pdb_chain$atom$b
```

```

# Plot B-factors with secondary structure annotation
plotb3(
  b_factors,
  sse = pdb_chain,
  typ = "l",
  ylab = "B-factor",
  xlab = "Residue Position",
  main = paste("B-factor Analysis for", pdb_id)
)
}

```

Example:

```
analyze_protein_drug_interaction("1E4Y")
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
 /var/folders/n0/stsdw7gn6q36btb46nvvr8hr0000gn/T/RtmpVW81zS/1E4Y.pdb exists.
 Skipping download

